

T024: STEAM VOIs as ROI Source

Status: planned Priority: high Created: 2026-05-14

Description

Load the prescribed STEAM spectroscopy voxels (tumor + contralateral) from the Bruker raw tree and expose them as an ROI source on both the per-ROI kinetics page and the parameter-map fitting page. The user gets a one-click, fully geometric ROI that matches exactly what the scanner prescribed — no manual drawing, no error from hand-traced contours.

Rationale

- Every study in this protocol acquires two STEAM voxels per subject: one in tumor, one contralateral.
- Those voxels are the canonical region the MR spectroscopy data was sampled from — analyzing T1 kinetics in the exact same region is the natural multimodal pairing.
- The contralateral voxel doubles as a built-in control for tumor kinetic parameters.
- Geometry comes for free from the Bruker method file (`PVM_VoxArrPosition / PVM_VoxArrSize / PVM_VoxArrGradOrient`); no segmentation needed.

Answered Questions

Question	Answer
Where does the STEAM geometry come from?	Bruker raw tree, <code><study>/<expno>/method, keys PVM_VoxArrPosition / PVM_VoxArrSize / PVM_VoxArrGradOrient</code>
How many VOIs per subject?	Always 2 (tumor + contralateral); list-typed schema keeps it extensible
Where does the sidecar JSON live?	<code>output/{dataset}/steam_voi.json</code> (per-subject; DICOM folder stays multi-subject)
Coordinate auto-fix?	Try identity + axis-flip combinations, pick best overlap with T1 brain mask, prompt to confirm before saving the chosen <code>scanner_to_patient 4x4</code>
Paired tumor-vs-contralateral plots?	Stretch (Phase 6); single-VOI flow ships first
Multi-subject DICOM/Bruker folders?	T1 combo selection auto-follows to the matching Bruker subject; auto-match by PatientID → StudyDate/Time → ordinal
Subject-link behavior?	Auto-follow on T1 combo change (matches 99% of workflows)

Open Questions

- [] Confirm `scanner_to_patient` defaults on this rig — once identified for one dataset, it should be the same across all subjects from the same scanner+coil configuration. May want a per-scanner preference, not per-dataset.
- [] Decide on filename for tumor/contralateral label-swap dialog state (in-memory only vs. persisted in JSON; persisting is simpler).

Data Model

Sidecar JSON

Location: `output/{dataset}/steam_voi.json`. One JSON per analyzed subject.

```
{
  "schema_version": 1,
  "subject_id": "B16",
  "scanner_to_patient": [[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1]],
  "voi": [
    {
      "label": "tumor",
      "method": "Bruker:STEAM",
      "frame": "bruker_scanner",
      "position_mm": [-2.5019681, -7.4787888, 0.8929790],
      "size_mm": [3.0, 3.0, 3.0],
      "orientation": [
        [ 0.66529950, 0.28308262, -0.69082618],
        [-0.34423973, 0.93740690, 0.05260525],
        [ 0.66247686, 0.20281157, 0.72110463]
      ],
      "source": {
        "kind": "bruker_method",
        "path": "../20250923_072207_Recht... B2_D16.../34/method",
        "keys_used": ["PVM_VoxArrPosition", "PVM_VoxArrSize",
"PVM_VoxArrGradOrient"]
      }
    },
    {
      "label": "contralateral",
      "method": "Bruker:STEAM",
      "...": "..."
    }
  ]
}
```

Schema notes:

- `frame` is the source coordinate frame of `position_mm/orientation` (`bruker_scanner` or `patient_lps`). Determines whether `scanner_to_patient` is applied during mask generation.
- `voi` is a list, not a scalar; ordering is irrelevant since each entry is labeled.
- `source` is bookkeeping; everything needed for analysis is captured in the numeric fields. Regeneration from `source.path` is supported if it exists.

Coordinate math

For each T1 voxel center p_{T1} in patient (LPS) coordinates, compute $q = \text{orientation} @ (\text{scanner_to_patient} @ p_{T1} - \text{position})$ and test $\text{all}(|q_i| \leq \text{size_mm}[i] / 2)$. Vectorized over the full 3D grid; sub-second for typical T1 volumes.

New module: `proxyl_analysis/steam_voi.py`

Function	Purpose
<code>parse_brucker_method(path) -> VoiDict</code>	Read PVM_VoxArrPosition/Size/GradOrient + ##\$Method from one method file
<code>scan_brucker_study(subject_root) -> list[VoiDict]</code>	Walk numbered subdirs of one subject folder, return all VOIs whose Method == Bruker:STEAM
<code>scan_brucker_root(root) -> dict[subject_id, list[VoiDict]]</code>	Walk every subject folder under a multi-subject root, return per-subject VOI lists
<code>load_voi_json(path) -> StudyVOIs</code>	Read sidecar JSON
<code>save_voi_json(path, study_vois)</code>	Write sidecar JSON
<code>voi_to_mask(voi, t1_geometry, scanner_to_patient) -> np.ndarray</code>	Rasterize one VOI to a 3D boolean mask in T1 voxel space
<code>voi_to_polygon(voi, z, t1_geometry, scanner_to_patient) -> np.ndarray</code>	Project one VOI onto one z-slice as a polygon for 2D overlay rendering
<code>auto_detect_transform(vois, t1_volume, t1_geometry) -> 4x4</code>	Try identity + axis-flip combinations, return best by overlap with T1 brain mask

UI Changes

`ui/main_menu.py` — **DicomScanResultsDialog**

Add a third selector row beside `t1_combo` / `t2_combo`:

```
T1 (PROXYL): [ Subject_B16 - PROXYL_T1 (series 12, 30 frames) ▼ ]
T2:          [ Subject_B16 - T2_TurboRARE (series 8, 25 slices) ▼ ]
STEAM:       [ Subject_B16 - 2 voxels (ExpNo 24, 32) ▼ ] [
Labels... ]
```

Plus a Bruker-tree path field at the dialog header level:

```
DICOM folder: /path/to/dicoms [Change...]
Bruker tree:  /path/to/20250923_..._B2_D16 [Change...] (optional)
```

Bruker tree path is remembered per-DICOM-folder in a small settings file so the user only picks it once per study series.

Auto-follow: when `t1_combo` changes, the STEAM combo jumps to the matching subject. Match cascade:

Priority	Match key	DICOM side	Bruker side
1	PatientID	tag (0010,0020)	<subject_root>/subject file
2	StudyDate + StudyTime (±2 hrs)	(0008,0020) / (0008,0030)	folder timestamp prefix
3	Ordinal position	series_number sort	folder name sort
4	Manual override	user selects from combo	

Visual hint: green check (priority 1), yellow "verify" (2/3), red "manual" (4).

Labels... button opens a small modal showing both voxels overlaid on the T1 mid-slice with a radio selection to swap which is tumor vs. contralateral. Default assignment uses laterality (configurable per-rig).

`ui/main_menu.py` — **MainMenuDialog ROI Analysis section**

Extend the existing ROI Source row:

```
ROI Source: ☐ T2   ☐ T1   ☒ STEAM VOI
           └─ when STEAM selected:
               Voxel: [ Tumor ▼ ]   [ Manage STEAM VOIs... ]
```

When `roi_source == "steam"`, hide the ROI Method row (Rectangle / Contour / Segment) — there's no drawing step.

A new status line under the ROI status: STEAM VOI: loaded (2 voxels) or STEAM VOI: not loaded.

`ui/fitting.py` — **KineticFitDialog**

No code changes for Phase 1–5; the dialog already accepts `roi_mask`. The upstream menu call computes the mask via `voi_to_mask(...)` and passes it in.

Phase 6 (stretch): when invoked with both VOIs, render a side-by-side "Tumor vs. Contralateral" panel using the same fit code run twice. Add a ratio panel for kinetic parameters.

`ui/parameter_map_options.py`

Two integration points:

1. **Fitting-region ROI** — extend the "ROI Processing" group:
2. ☒ Process whole image

3. ◦ Process within ROI only
4. ROI: ◦ Reuse existing ◦ Draw new ◦ Use STEAM VOI [Tumor ▼]

Bounds pixel-level fitting to the prescribed voxel. Much faster than whole-image fitting.

5. **Measurement ROI** — add a sibling button to the existing measurement-ROI flow: "Measure within STEAM VOI...". Picks tumor or contralateral and computes stats from the 3D mask. Modify `_save_pm_metric_bundle` to accept a label string (e.g. `STEAM_Tumor`) so output filenames are self-identifying instead of just integer-indexed.

ui/steam_voi.py (new) — STEAM VOI dialog

Standalone modal for the "Manage STEAM VOIs..." / "Load STEAM VOI..." actions. Reuses ProxyFit styling. Sections:

- Source picker (Bruker study folder vs. single method file)
- Found-VOIs table (Use / ExpNo / position / label)
- Overlay preview on T1 mid-slice
- Scanner→patient transform editor (auto-detect / manual 4×4)
- Save & Use button

Persistence

Artifact	Location	Lifetime
<code>steam_voi.json</code>	<code>output/{dataset}/</code>	Permanent; per-analyzed-subject
Cached masks	<code>output/{dataset}/steam_voi/{label}_mask.npz</code>	Regenerated if missing
Bruker-tree path mapping	<code>output/.steam_voi_bruker_paths.json</code>	Per-DICOM-folder; remembers picker choice
<code>scanner_to_patient 4×4</code>	inside <code>steam_voi.json</code>	Per-dataset; once tuned, never re-touched

JSON is the source of truth; everything else is a cache.

Implementation Phases

Phase	Scope	Done when
1. Core math + parser	<code>steam_voi.py</code> + unit tests	<code>voi_to_mask</code> round-trips through a synthetic T1 geometry; Bruker

Phase	Scope	Done when
2. Loader dialog	ui/steam_voi.py + folder scan	method parser matches the values extracted by hand from 34/method Pointing at 20250923_... auto-finds both STEAM voxels, writes JSON, shows preview
3. Menu integration	DicomScanResultsDialog STEAM combo + auto-follow; MainMenuDialog ROI source option	"STEAM VOI → Tumor" runs kinetic fit on tumor VOI mask
4. Parameter-map fitting ROI	parameter_map_options.py	Fit-within-VOI runs and produces correctly-bounded parameter maps "Measure within STEAM VOI" produces summary bundle with STEAM_Tumor / STEAM_Contralateral labels
5. Measurement-ROI integration	parameter_map_options.py + fitting.py summary bundle	
6. Stretch: paired tumor vs. contralateral	KineticFitDialog	Side-by-side panel + ratio metrics

Acceptance Criteria

- [] Loading a Bruker study root populates output/{dataset}/steam_voi.json with both VOIs, no manual editing required
- [] Both VOIs render correctly as overlay boxes on the T1 (visually verified)
- [] Kinetic fit on STEAM tumor VOI matches kinetic fit on a hand-drawn ROI of the same region within tolerance
- [] Parameter-map fitting bounded by STEAM VOI runs in a small fraction of the time of whole-image fitting
- [] Session reopens without re-loading the Bruker tree (JSON sidecar is the only requirement)
- [] Multi-subject DICOM folders: T1 combo selection auto-follows STEAM combo to the matching subject by PatientID / StudyDate / ordinal
- [] scanner_to_patient auto-detect succeeds on the reference dataset (20250923_..._B2_D16_...); manual override path documented

Files Affected

```

proxyl_analysis/steam_voi.py           # NEW: parser + math + persistence
proxyl_analysis/ui/steam_voi.py        # NEW: loader dialog
proxyl_analysis/ui/main_menu.py        # STEAM combo in
DicomScanResultsDialog;                # STEAM source option in
MainMenuDialog
proxyl_analysis/ui/parameter_map_options.py # STEAM as fitting ROI +
measurement ROI

```

```
proxyl_analysis/ui/fitting.py
panel
proxyl_analysis/io.py
load
tests/test_steam_voi.py
JSON
docs/changelog.md
```

```
# Phase 6: paired tumor/contralateral
# Discover steam_voi.json on session
# NEW: unit tests for parser, math,
# Version entry
```